



Inspiring Excellence

Lecture Plan

CSE331: Automata And Computability

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Lecture 1: Theory of Computation

- Computability and Complexity Theory
- Automata Theory
- Preliminaries:
 - Sets and Set Operations
 - Proofs

Lecture 2: Deterministic Finite Automata

- Alphabets, Strings, Languages
- Finite Automata (FA)
- Deterministic Finite Automata (DFA)
 - State Diagram
 - Formal Definition
 - Designing DFA
- Regular Language: A language is called a **Regular Language** if some finite automaton recognizes it.

Lecture 3: DFA (Continued)

- DFA Negation
- Cross Product of DFA
- Regular Operations
 - Union
 - Concatenation
 - Kleene Star
- Closure of RL Under Union, Intersection, Complement

Lecture 4: Non-deterministic Finite Automata

- Non-deterministic Finite Automata (NFA)
 - State Diagram
 - Formal Definition
 - Designing NFA
- Difference Between DFA and NFA

Lecture 5: Equivalence of NFA and DFA

- Epsilon Closure
- NFA to DFA using the Subset Construction
- Theorem: A language is regular **if and only if** some NFA recognizes it.

Lecture 6: Regular Expressions

- Regular Expressions (RE)
 - Formal Definition
 - Designing RE
- Precedence of Regular Operators
- Union & Concatenation of RE with ε , $\emptyset$

Lecture 7: RE to NFA

- Closure of RL Under Concatenation, Kleene Star
- RE to NFA using Thompson's Construction Method
- Lemma: If a language is described by a regular expression, then it is regular.

Lecture 8: FA to RE

- GNFA
- FA to RE using State Elimination Method
 - Steps of the Conversion
- Theorem: A language is regular **if and only if** some regular expression describes it.

Lecture 9: Pumping Lemma

- Pumping Lemma for RL
- Proving Non-regularity using Pumping Lemma
 - Steps of Proof
 - Using Closure Properties of RL
 - Pumping Up and Pumping Down
- Necessary Condition for Regular Languages

Lecture 10: Context-Free Grammar

- Non-regular Languages
- Context-Free Grammar (CFG)
 - Formal Definition
 - Derivation & Parse Tree
 - Designing CFG
- Context-Free Languages

Lecture 11: CFG (Continued)

- DFA, NFA to CFG
- RE to CFG
- RL is a subset of CFL
- Closure of CFL Under Union, Concatenation, Kleene Star

Lecture 12: Ambiguity of CFG

- Leftmost and Rightmost Derivation
- Ambiguity of CFG
- Removing Ambiguity

Lecture 13 & 14: Pushdown Automata

- Pushdown Automata (PDA)
 - State Diagram
 - Formal Definition
 - Designing PDA
- PDA for RL
- Additional Topics:
 - Equivalence of CFG and PDA
 - Pumping Lemma for CFL

Lecture 15: Turing Machine

- Turing Machine (TM)
 - Formal Definition
 - State Diagram
 - Designing TM
- Difference with FA

Lecture 16: TM (Continued)

- Church-Turing Thesis
- Turing Decidable and Recognizable Languages
- Proving a Language Decidable
 - Notations for Encoding
- Universal Computing Machines
- The Halting Problem
 - Hilbert's Tenth Problem



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The End